Examiner only

SECTION B

*Answer **all** questions in the spaces provided.*

8. A student was told that he could prepare chloroethane, C_2H_5Cl, by mixing ethane with chlorine. He added 2.0 g of ethane to excess chlorine and left the mixture exposed to ultraviolet light for several hours. He was then able to use a university laboratory to see whether chloroethane had been made.

(a) State an instrumental method by which the sample could be analysed. Explain how this would show that chlorination had occurred. [2]

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(b) State why it is necessary to use ultraviolet light when making chloroethane from ethane. Give equations to show the mechanism for the formation of chloroethane. [4]

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(c) The student found that he had made 1.0 g of chloroethane. Calculate his percentage yield. [3]

Percentage yield = ... %

(d) The student was disappointed by the yield he obtained but his teacher told him that the yield was always poor due to other products being formed in this reaction.

Identify **two** other organic products, apart from chloroethane, and explain how they are formed. [2]

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